A meta-analytic predictive approach to fitness-related dropout rates of special operations forces candidates

Alain Dössegger, MSc[1,2]; Thomas Gsponer, PhD[3]; Christian Protte, MD[1]; Thomas Wyss, PhD[1]; Rahel Gilgen-Ammann, PhD[1]; LTC Denis Bron, MD[4]; 1LT Marina Veil, MSc[5]; Martin Flück, PhD[1,6]; Markus Gerber, PhD[2]; Zeno Stanga, MD[7,8]

ABSTRACT

Introduction:
High dropout rates during the selection of Special Operations Forces (SOF) are well-documented and often linked to inadequate physical preparedness and subsequent injuries. Training recommendations in a preparatory selection phase may offer a strategy to reduce dropout.

Objective:
To assess whether fitness-related dropout was reduced in a study population of SOF candidates receiving training recommendations after fitness assessment, compared to historical attrition rates.

Methods:
Using a prospective observational design, male candidates across Swiss Armed Forces and police SOF units received individualized training recommendations in the preparation phase of the main selection period. Dropout due to inadequate physical preparedness for the demands of SOF selection and basic training was tracked through all selection stages. A Bayesian analysis compared observed fitness-related dropout rates with a historical control group using a Meta-Analytic Predictive approach.

Results:
Fitness-related dropout occurred in 29% of participants (40 of 137), compared to a historical average of 42%, corresponding to a relative reduction of over 30%. The posterior probability of a reduced dropout rate (odds ratio <1) was 91.6%.

Conclusion:
Targeted training guidance as part of human performance management may help to reduce fitness-related dropout rates among SOF candidates.

[1]Swiss Federal Institute of Sport Magglingen SFISM, Magglingen 2532, Switzerland
[2]Department of Sport, Exercise and Health, University of Basel, Basel 4052, Switzerland
[3]Stadl Partners GmbH, Münsingen 3110, Switzerland
[4]Faculty at Aeromedical Institute, Swiss Aeromedical Center, Swiss Air Force, Dübendorf 8600, Switzerland
[5]Armed Forces Personnel, Swiss Armed Forces, Bern 3003, Switzerland
[6]Department of Endocrinology, Metabolism and Cardiovascular System, University of Fribourg, Fribourg 1700, Switzerland
[7]Centre of Competence for Military and Disaster Medicine, Swiss Armed Forces, Bern 3003, Switzerland
[8]Division of Diabetes, Endocrinology, Nutritional Medicine and Metabolism, University and University Hospital of Bern, Bern 3010, Switzerland

The views expressed in this material are those of the authors, and do not reflect the official policy or position of the U.S. Government, the Department of Defense, or the Department of the Army.

Corresponding author: Alain Dössegger, MSc, Switzerland (alain.doessegger@baspo.admin.ch).

doi:https://doi.org/10.1093/milmed/usaf619

© The Association of Military Surgeons of the United States 2026.

This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (https://creativecommons.org/licenses/by-nc/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact journals.permissions@oup.com

INTRODUCTION

Selection for Special Operation Forces (SOF) are designed to find the most suitable candidates, by pushing them to their physical and psychological limits and beyond. Overall dropout rates are as high as 71.5% in Dutch[1] or even around 80% in Australian, U.S. Navy, and Belgian Tier 1 SOF (elite, strategically tasked units with the highest resources and secrecy, e.g., U.S. Delta Force, Austrian Jagdkommando, UK SAS).[2–4] Other sources report attrition rates from 50% upward.[5] In Swiss Tier 1 SOF, detailed dropout rates are classified information. For military Tier 2 SOF (special forces units with lower budgets compared to Tier 1), overall dropout rates range from 59% to 87%, and in police Tier 2 SOF from 23% to 70%.[6]

Reasons for dropout are multifaceted. According to personal communication with recruiting authorities from the Swiss Armed Forces and cantonal police units, dropout reasons include mismatch of fitness to the progressively higher physical demands, loss of motivation, injuries, sleep deprivation issues, non-bearing of psychological pressure, just to name a few. In line with this, numbers from the United States suggest that self-efficacy, cognitive aptitude, voluntary request to leave selection and lack of physical fitness are the predominant reasons for attrition.[7,8]

Given that every dropout represents a significant loss of invested personal and institutional resources, structured pre-screening of both physical and psychological readiness is increasingly recognized as a necessary step prior to SOF qualification.[2] For instance, the Occupational Physical Assessment Test led to around 10% overall reduction in attrition in the U.S. Army, saving $50 million a year.[9] This general U.S. Army reference illustrates the potential of systematic screening. SOF-specific data remain inaccessible. However, artificial intelligence (AI)-scratched numbers from internet sources and personal communication from the SOF community roughly suggest $600,000 to over $1 million is spent per deployable SOF operator, illustrating the potential financial losses if candidates drop out due to preventable fitness issues at later stages of the SOF selection process.

The goal of a selection process is to de-select those who might get injured or give up too early during basic training, or, the other way around, to detect those who are strong and resilient enough to pass through all the trainings and selection stages and become a valuable member of the SOF family. Selection criteria for SOF may comprise objective and subjective elements. For example, minimal time-to-task (e.g., 2,000-3,000 m run), maximal strength (e.g., bench presses) or repetitions within 2 minutes (e.g., push-ups and pull-ups), are considered objective fitness selection criteria. Candidates' behavior in a certain (team) setting under pressure, assessments of motivation, intelligence, personality, and resilience, as well as the warrant officer's gut feeling of physical and mental preparedness for the basic training are considered subjective selection criteria. The selection criteria are based on the personal experience of the recruiting authorities about what it takes to be an operator. Recruiting authorities consist of warrant officers, long-serving SOF with a physical fitness background, and physicians consulting them.

One key reason for de-selection is a lack of physical preparedness. Drop rate due to physical performance in U.S. green beret is estimated at 20%.[8] Low physical fitness increases the relative risk for musculoskeletal injuries in the military setting by a factor of 2.34.[10] Of the 800 soldiers who participated in a study within the U.S. Army Special Forces Assessment and Selection Course, 38% experienced one or more injuries.[11] SOF trainees conducting qualification training had up to 68% injured in a training period.[12] Further, SOF lose a significant amount of lean body mass tied to energy deficits and sleep deprivation during training,[13] making physical preparedness and resilience key to withstand the physical challenges during the basic training phase.

Structured physical training interventions prior to selection can significantly reduce attrition and improve physical readiness. Wenger et al.[4] highlight that structured preparatory courses have led to improved candidate readiness and performance. These enhancements are expected to contribute to lower attrition rates. Following the introduction of the Special Warfare Human Performance Support Group, musculoskeletal injury rates during the apprentice course declined from 2.4 to 1.0 cases per 100 trainee-weeks—a nearly 20% reduction in cumulative incidence compared to historical data.[15] In the Guardian Angel/Special Tactics Assessment and Selection course, structured feedback on individual strengths and weaknesses was also a key component.[16] Likewise, the integration of athletic trainers in U.S. Air Force basic military training led to 25% fewer musculoskeletal injuries and a 15% reduction in all-cause attrition, achieved through, for example, individualized gait analysis and tailored training interventions.[17] These findings support the use of targeted and individualized training strategies in high-performance military settings, as advocated by Pattyn et al.[2]

Individualization can be attained in different ways. For example, aerobic endurance training prescription based on individual ventilatory thresholds results in better gains than standardized training zone recommendations, for example, based on heart rate reserve.[18] Further, specific genetic variations (polymorphisms), such as in the *ACE*, *ACTN3*, *MCT1*, *CKM*, *PTK2*, and *HIF1A* genes—are known to be associated with different physiological responses to different types of exercise stimuli.[19–24] Understanding the associations may inform individualized training strategies. Interindividual variability in training responses and corresponding exercise-induced gene expression patterns in skeletal muscle indicate a genetic component to training responsiveness.[25–29] A resistance training program by Jones et al.[30] which matched participants' individual genotype showed greater improvements in countermovement jump and aerobic performance than mismatched programs.

Beyond physiological improvements, individualized feedback may also strengthen candidates' motivation and follow-through. The Sport Commitment Model identifies enjoyment, personal investment, opportunity, social support, and limited alternatives as key drivers of sustained engagement.[31] Individualized training recommendations may transform selection from a high-stakes screening into a developmental opportunity. Moreover, such feedback may boost candidates' physical self-concept, which is closely tied to motivation.[32] Feeling capable of improvement can foster a sense of progress and control, increasing the motivation to train. Thus, feedback from pre-selection assessments that include tailored recommendations may create a motivational "attention effect."

Given this background, the purpose of this study was to examine whether individualized feedback and training recommendations from an additional pre-selection fitness test would reduce dropout due to inadequate physical preparedness for the demands of SOF selection and basic training of SOF candidates in Swiss Armed Forces and police SOF. The researchers hypothesized that pre-selection fitness tests with individualized feedback reduce fitness-related dropout during the SOF selection by 50%.

METHODS

The prospective observational study P(O|TS)[33] (Probability of becoming an Operator given specific Test resultS) recruited candidates applying to various Swiss SOF Tier 1 and 2 units,

including para-reconnaissance units, military police SOF, police SOF, and grenadiers. Participation in the study was voluntary. Candidates were assessed in parallel with each unit's standard multi-phase selection process assessments that started in 2023 or 2024. Participant recruitment details are described in detail elsewhere.[33] In short, candidates were asked for their consent by the respective recruiting organization prior to the start of the main formal selection phase, but after initial pre-screening by the different units. Candidates who agreed to participate and signed an informed consent form were pseudonymized by a unique study ID by the recruiting organization, their personal details were not disclosed to the study team. Only candidates that passed the initial unit-specific medical and formal pre-screening, as well as the Physical Activity Readiness Questionnaire assessing medical contraindications for exercise, and agreed to participate, were included as study participants.

Participating candidates completed a pre-selection physical fitness assessment, consisting of multiple fitness tests designed to measure SOF-specific fitness requirements. This included endurance (e.g., VO2max, performance at VT1), tested by cardiopulmonary exercise testing (CPET) on a treadmill, maximal isometric and dynamic strength (e.g., maximal deadlifting strength), striking and jumping power, balance and stability, and executive function (measured via Vienna Test system and a CQB-like reaction task with a FX-handgun), as well as health and training questionnaires. All tests are described in detail in Dössegger et al.[33] Upon agreement to an additional informed consent on the collection of genetic polymorphisms, candidates' mucosa was sampled to assess fitness-related genotypes via polymerase chain reaction.

Following the study fitness assessment, test results and scores (z-normalized test results for comparison with the study sample) were reported to each participant via a password-protected webpage using their study ID as identifier. SOF-tailored training plans, individual aerobic endurance training zones as well as individualized training stimuli recommendations were provided: A downloadable 3-month tailored training plan for either (military or police) grenadiers or para-reconnaissance units was developed by training experts, aiming at the development of all fitness components while focusing either on strength or on weight-bearing aerobic endurance (**Tables S1 and S2**; see online supplementary material). Based on CPET, the individual endurance training zones were communicated. Genotypic information was used to recommend training stimuli that could be added to the basic training (**Table S3**; see online supplementary material), including a short explanation on the stimuli, via the beta-version of the feedback platform.

Implementation of these training recommendations was not supervised or monitored and was left to the discretion of the participants. Fitness test results were not disclosed to the selection authorities. Neither study participation nor the test results of the study influenced the selection/de-selection of candidates; these decisions were only taken inside the selecting units. **Figure S1** (see online supplementary material for a color version of this figure) illustrates the process of the P(O|TS) study.

Primary Endpoint Definition

The primary endpoint was fitness-related dropout from the standard, unit-specific selection process. We restricted the endpoint to: dropout due to overuse and/or injuries that are caused by inadequate physical preparedness, by the end of all selection and training courses (excluding dropouts due to motivation or other non-physical fitness related issues, excluding rolled candidates due to illness). The selection authorities of the respective organizations evaluated whether each candidate that participated in this study was de-selected due to physical fitness limitations. This attribution was uniquely based on their point of view and was assessed using a structured evaluation questionnaire with the following predefined response options:

1. Dropout was primarily caused by insufficient physical fitness.
2. Dropout was partially caused by insufficient physical fitness.
3. Dropout was unrelated to physical fitness deficiencies.

Based on these responses, a binary endpoint variable was derived. Candidates were coded as 1 (yes) if their dropout was either primarily or partially attributed to insufficient physical fitness (i.e., responses 1 or 2), and as 0 (no) if they either completed the selection process or if their dropout was unrelated to physical fitness (response 3). The primary endpoint was reported to the project leader using the study ID. If only a binary answer (yes or no) was reproducible, the answers were coded accordingly.

Study Objective and Bayesian Decision Criterion

The goal of our analysis was to evaluate in a probabilistic way whether the study group's dropout rate due to overuse and/or injuries, that are caused by inadequate physical preparedness, was lower than the one observed historically. The historical control group was defined as candidates from selections to the same or similar SOF units in the past without training recommendations. In addition, the researchers compared the *overall* SOF-dropout rate in the study years candidate's cohort with the overall dropout rate of the cohorts in previous years.

Technically, a Bayesian approach was chosen: Let θ_s represent the fitness-related dropout rate among candidates following the protocol of this study, and θ^* represent the fitness-related dropout rate in the historical control group. To determine whether the candidates participating in the study had reduced fitness-related dropout, the researchers calculate the odds ratio (OR) comparing the two groups. The odds ratio is defined as:

$$OR\left(\theta_s, \theta^*\right) = \frac{\theta_s\left(1-\theta^*\right)}{\left(1-\theta_s\right)\theta^*}$$

The researchers defined success as the probability of our participants having a smaller OR for fitness-related dropout compared to the historical cohort being larger than 80%:

$$P\left[\mathrm{OR}\left(\theta_s,\theta^*\right)<1\right]>0.8$$

and that the probability that the OR is less than 0.5 is larger than 50%:

$$P\left[\mathrm{OR}\left(\theta_s,\theta^*\right)<0.5\right]>0.5$$

Historical Data

Historical fitness-related dropout data from two Swiss Armed Forces and three police SOF units were obtained via personal communication with the respective selection authorities of the different units. Reported fitness-related dropout rates range from 0.32 to 0.51 (corresponding to 32%-51%) in the years prior to the study. Historical fitness-related dropouts include selection-dropouts attributable to inadequate physical preparedness for the demands of SOF selection and basic training only, ensuring alignment with the study's primary endpoint. All information was sourced from the same selection authorities involved in the current study and was collected at the outset of the project.

Specifically, military Tier 2 units provided dropout data from selection cycles conducted from 2019 to 2022 (typically two rounds per year), while the Tier 1 military unit contributed aggregated dropout figures dating back to 2018. Tier 1 police SOF reported averaged dropout rates from selection rounds conducted during the years prior to the study, that is, 2018-2021. One Tier 2 police unit provided data from 2013, 2016, 2019, and 2023, another unit provided data from 2017 and 2019. Although selection authorities were asked to provide exact figures, classification constraints limited data precision.

Importantly, no changes were made to the selection procedures in the participating units during the transition from the historical to the present study period. This consistency allowed for comparisons between historical dropout rates and those observed during the present study phase.

To synthesize the heterogeneous historical data across units while accounting for between-unit variability, a Bayesian hierarchical meta-analysis was performed on these rates (**Figure 1**), assuming exchangeability between units—that is, the dropout rate is considered similar across different SOF units. The resulting posterior predictive distribution has a median of 0.42 and a 95% predictive interval of 0.23-0.61, corresponding to an estimated dropout rate of 42%. This information was used as the prior distribution representing the control group, and corresponds to the Meta-Analytic Predictive (MAP) prior of Neuenschwander et al.[34] In addition, a robust MAP prior was constructed by mixing the MAP (80%) with a uniform distribution containing no prior belief (20%). This historically informed prior represents our belief about the dropout rate for

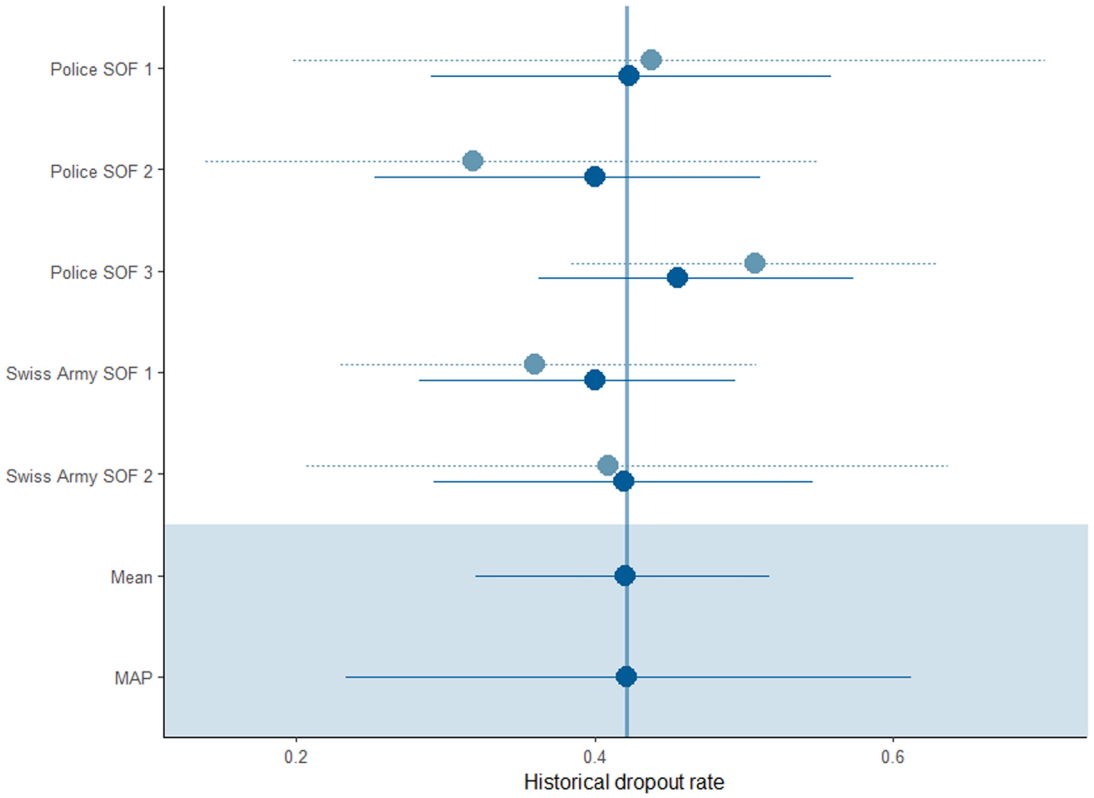


Figure 1. Forest plot of historical dropout rates and Meta-Analytic Predictive (MAP) prior construction. Each row represents a special operations force (SOF) unit with its observed dropout rate (light blue) and the corresponding model-based estimate (dark blue) with 95% credible intervals. Estimates were generated using a hierarchical Bayesian model (gMAP). The bottom rows show the posterior mean across all groups ("Mean") and the predictive MAP prior ("MAP") for the study cohort. Estimated between-group heterogeneity (τ) was moderate at 0.26 (95% CrI: 0.01-0.75), supporting the use of dynamic borrowing.

the current selection procedures of the respective units (without pre-selection fitness tests).

The information in the MAP can be considered equivalent to a number of observations, which is called the effective sample size (ESS) of the prior.[35] The ESS based on the MAP distribution was 46, the information in the robustified MAP corresponds to 32 individuals. In other words, the information corresponding to the MAP distribution is equivalent to 32-46 candidates.

Study Prior

The study prior reflects the anticipated fitness-related dropout rate among SOF candidates undergoing the study pre-selection in addition to the usual selection procedures individually defined by the respective units. The prior has a median of 0.42, which to the estimated dropout rate in the historical control group. The 95% credible interval (0.13-0.84) of the study prior expectation reflects substantial uncertainty. This weakly informative prior was chosen to allow the data to dominate the posterior inference, while still anchoring expectations in plausible values. The researchers thus assume an a prior 50:50 chance that the dropout rate in the study population is higher or lower than in the historical control group.

Following the Bayesian probability updating procedure, the study prior was then updated with observed fitness-related dropout data to yield the posterior distribution of the fitness-related dropout rate among candidates following the present study protocol θ_s. This posterior distribution was then compared with the MAP prior from the fitness-related dropout rate in the historical control group θ^* to compute the defined success criteria.

All study data were collected and managed using REDCap electronic data capture tools (v13.7.3)[36] hosted at the Federal office of sport FOSPO, Magglingen, Switzerland.

The study was performed in accordance with current ethical guidelines (Declaration of Helsinki, as revised in 2013) and was approved by the Human Research Ethics Committee of the Canton of Berne (approval no. 2022-00767), as well as the research committee of the Medical Services Directorate of the Swiss Armed Forces.

RESULTS

A total of 175 male SOF candidates were evaluated with the study's fitness test battery. Complete endpoint data were available for 137 candidates from four different units (65 from the Swiss Armed Forces and 72 from police SOF). The researchers excluded candidates from one unit that raised internal fitness requirements with the start of the study ($n=12$), which would have compromised comparability with historical data. They also excluded candidates from units with no historical data ($n=14$). One unit lost the key list with the candidate's study ID, so no dropout data could be obtained ($n=9$). Three candidates did not finish basic training but did not drop out, instead they will reenter the basic training course this year. Anthropometric data of the study participants are summarized in **Table 1**. No between-unit differences were denoted. All candidates agreed to the genotyping analyses.

Table 1. Characteristics of Participating SOF Candidates ($n=140$)

	Mean ± SD
Age (years)	25.6±6.1
Height (cm)	179.7±6.6
Body mass (kg)	78.6±9.8
BMI (kg/m²)	24.3±2.4
Total self reported training hours per week (h)	7.3±3.6

In total, $n=40$ SOF candidates dropped out from the unit-specific selection process due to physical fitness limitations.

After incorporating the observed dropout data from our study in the Bayesian updating procedure, the posterior distribution for the dropout rate had a mean of 0.294 ($SD=0.039$), with a median of 0.293 and a 95% credible interval ranging from 0.221 to 0.372. This corresponds to a fitness-related dropout rate of ∼29% (95% CrI [22.1-37.2%]).

The probability that this rate is lower than the historical dropout rate is 0.916 (∼91%). The probability that the new dropout rate is below 26.4% (i.e., OR < 0.5), was 0.127 (∼13%). **Figure 2** illustrates the shift of the fitness-related dropout rate in this study (red curve) relative to the historical dropout rate (gray/black curves).

Overall dropout rate (all dropout reasons considered) in the years prior to the P(O|TS) study (until 2023) ranged from 32% to 85% in military Tier 2 units. In the study years (2023-2025), overall dropout rate ranged from 28% to 87%. The overall dropout rate among candidates who participated in the study in addition to the regular selection process was 42% (**Table 2**). No information about the dropout rate of non-study participants during the study period was available.

DISCUSSION

This study aimed to evaluate whether fitness-related dropout rate in Swiss SOF candidates is reduced after candidates received feedback, attention, and individualized training recommendations. The researchers hypothesized that the probability of our participants having a smaller OR for fitness-related dropout compared to the historical cohort was larger than 80%, which we confirmed. The observed 29% fitness-related dropout rate suggests a high probability (91%) for a reduced dropout rate in study participants compared to historical expectations of 42%. In contrast, they could not confirm the second hypothesis. The probability that the dropout rate was half as large compared to the historical fitness-related rate was only around 13%. Still, the reduction of approximately 13 percentage points corresponds to a relative reduction of over 30%.

The use of a Bayesian hierarchical model allowed the researchers to integrate heterogeneous historical data across units and estimate an MAP prior, as a proxy for what dropout would have been without the present study conducted in parallel to the unit's selection process, that is, a model-based counterfactual.

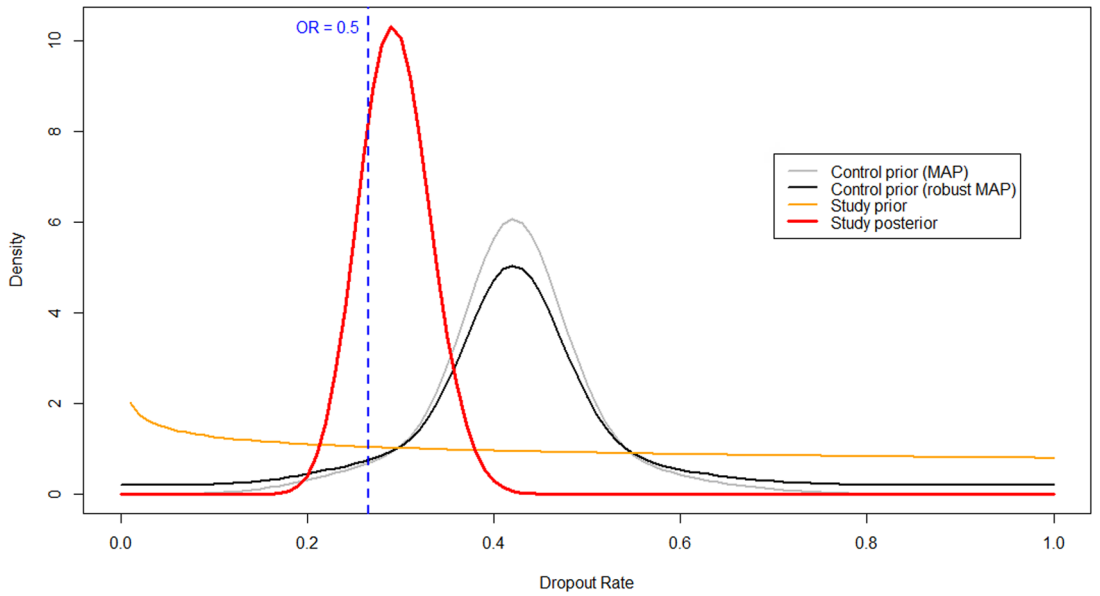


Figure 2. Posterior distribution of the fitness-related dropout rate in the study group (red), compared to the Meta-Analytic Predictive (MAP) prior derived from historical data (gray), and a robustified MAP prior (black). The posterior dropout rate, based on 40 fitness-related dropouts among 140 candidates (mean = 29%), is less than the historical dropout rate with a probability of 92%, indicating strong evidence for a reduction in fitness-related dropout relative to historical expectations. The blue dashed line indicates half the odds (0.36, 26.6% dropouts) of the historical control (0.724, 42% dropouts). The study prior (orange) is the expected (non-)change of fitness-related dropout in the study group compared to historical cohorts.

Table 2. Overall Dropout Rates (All Reasons) in Different SOF Units over the Time Course

	2013	2016	2017	2018	2019	2020	2021	2022	2023	2024
Tier 1 SOF				73%					70%	
Tier 1 SOF				87%						«unchanged»
Tier 2 SOF				41%	32%	39%	42%	48%	28%	
Tier 2 SOF						69%	85%	59%	87%	
Tier 2 SOF	64%	72%			69%		63%		67%	
Tier 2 SOF			71%		50%					
P(O|TS) study									42%	

The observed reduction in the fitness-related dropout rate seems to be noticeable, as overall dropout rates did not markedly change when evaluating the statistical data available.

Even though training was left to the discretion of candidates, it seems plausible that candidates may have had a benefit from participation in the study. The researchers hypothesize that simply providing individualized, evidence-based training guidance based on transparent fitness performance may be sufficient to yield better physical adaptations to selection courses in a highly motivated tactical athlete population. They advocate that in this population, a fitness test battery should not only be used as selection instrument but instead serve as individualized support instrument. Combining the analysis of the current SOF-specific fitness performance with the genotype-informed training susceptibility (the potential to develop) and individualized feedback has been successfully implemented in Swiss SOF. The test battery, validly assessing power, strength, endurance, stability, and reaction time, takes around 1.5 hours per candidate. With some restrictions in precision, e.g., estimating maximal aerobic endurance capacity instead of performing CPET with every candidate, and with a circuit-like test organization, a considerable high number of candidates could be tested per day. For genotyping, establishing a safe and regulatory-conform pipeline with a certified laboratory is key. The researchers' secure, individualized feedback to the candidates is already used in some SOF units and may be adapted to other selecting authorities.

High attrition rates during SOF selection and basic training are a well-documented issue.[2,4] These losses carry significant but preventable financial costs, particularly when dropout is driven by inadequate physical preparedness for the demands of SOF selection, such as load-bearing marches, or strength resilience. Our findings reinforce a broad and growing body of evidence showing that early identification of physical readiness and targeted support can significantly reduce injuries and dropout, making SOF selection more efficient and cost-effective.[9,15,17]

The effect occurred without intensive supervision or mandatory training structures, contrasting favorably with other studies where support programs were tightly integrated. Thus, the researchers' approach seems to have a scalable potential in both military and police SOF settings.

A likely mechanism behind the observed reduction is an increase in self-efficacy among candidates. According to Bandura,[37] self-efficacy is strengthened through different sources, from which we likely addressed at least two: verbal persuasion via individualized feedback and training recommendations, and reduced uncertainty about capabilities via the provision of an individual strengths and weaknesses fitness profile. Notably, self-efficacy and motivation were identified as key predictors of dropout in Dutch SOF.[38]

Several limitations should be mentioned: First, selection bias is a threat to interpreting our study results. Study participation was voluntary, so candidates with higher motivation or better baseline fitness may have been more likely to opt in, which could partially explain the observed lower overall dropout rate in the study cohort compared with the overall dropout rate in the same years. However, this risk is mitigated by the fact that participation of all candidates was strongly encouraged by the units themselves, and the sample includes candidates from multiple SOF units with varying baseline dropout rates. Second, it was not possible to recruit a concurrent control group due to ethical and organizational reasons, hence the researchers relied on historical dropout data instead. Third, the MAP prior assumes exchangeability between SOF units, potentially overlooking meaningful structural or cultural differences in dropout drivers in different SOF units. Fourth, as for the endpoints, the researchers relied on personal communication with the medical specialists and warrant officers in the units. However, no hard decision criteria were available. It seems noteworthy that the number of selected candidates is not solely determined by candidate readiness. Promotion depends also on the specific operational needs and financial resources of each unit. Officers across units reported that these constraints remained stable during the study period, and that promotions closely followed the number of retirements or vacancies. Crucially, no candidate was promoted unless they met al. internal selection criteria. In cases where more candidates were deemed eligible than available positions, some units maintained a waiting pool and postponed new selection rounds until that pool was exhausted. These limitations should be kept in mind when interpreting the practical impact of reduced dropout.

A key strength of this study is the use of a Bayesian hierarchical meta-analysis to synthesize historical dropout data into an MAP prior. This allowed for probabilistic comparisons with the study group while formally accounting for uncertainty and between-unit variability. The Bayesian decision criterion enabled practically meaningful interpretation. This allowed for a transparent and quantitatively grounded interpretation of success, distinguishing our approach from retrospective or anecdotal analyses typically seen in the tactical settings. Further, the provision of training plans and even individualized recommendations in a secure feedback pipeline, as well as the prospective study design are further strengths of this research. Finally, the researchers were able to obtain endpoints while keeping the anonymization of the SOF candidates intact.

The results align with this international trend toward predictive, data-driven planning in SOF human performance management.[39] However, the effectiveness and efficiency of such approaches needs to be evaluated in different settings.

CONCLUSIONS

This study suggests that an SOF-specific pre-selection fitness screening, combined with individualized feedback and training recommendations, may meaningfully contribute to mitigate fitness-related dropout in Swiss SOF candidates. The observed 29% dropout rate was substantially lower than the historical benchmark of 42%, with a high posterior probability of improvement based on a robust Bayesian meta-analytic model. While the study was observational and subject to possible selection bias, the consistency of results across units and the unsupervised nature of the intervention point to the practical feasibility and scalability of this approach. Beyond identifying readiness, such assessments may enhance candidates' self-efficacy and motivation. Both are known to influence attrition in SOF training. Conducting these pre-selection tests prior to formal selection would allow candidates to engage in targeted physical preparation aligned with the demands of their intended SOF unit. Future evaluations should include a prospective, controlled experimental design comparing cohorts with and without the pre-selection procedure, with supervised training implementation, to elucidate intervention effects.

ACKNOWLEDGMENTS

The authors would like to thank: Beuchat, Ch., Braihier, P., Droz, G., Eddie, Faude, O., Fischer, R., Finch, Gimli, Goldhorn, A., Grob, T., Gunziger, D., Hägi, G., Hänni, M., Hedinger, B., Hollenstein, M., Keller, T.A., Küffer, A und D., Kurz, T., Mutti, S., Quadrio, K., Rappo, T., Rüegsegger, A., Stoll, D., Veil, M., Züger, R., Züllig, S., for recruiting SOF operators, and assistance during testing. Thanks to Gimli, Finch, Berd, Eddie, and Züllig, S., for providing insightful information about SOF critical success factors. The authors acknowledge the support of the Centre of Competence for Military and Disaster Medicine, the Special Forces Command (SFC) of the Swiss Armed Forces, as well as the Faculty at Aeromedical Institute.

The authors acknowledge the use of AI-assisted language tools, including ChatGPT, for refining certain sections.

SUPPLEMENTARY MATERIAL

Supplementary material is available at *Military Medicine* online.

FUNDING

This research received funding from the Swiss Armed Forces Innovation Board.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY

The data that support the findings of this study are available on reasonable request from the corresponding author.

INSTITUTIONAL ANIMAL CARE AND USE COMMITTEE

Not applicable.

INSTITUTIONAL CLEARANCE

Not applicable.

INDIVIDUAL AUTHOR CONTRIBUTION STATEMENT

Conceptualization, A.D., T.G, C.P.; methodology, A.D., C.P., M.G., M.F.; formal analysis, A.D., T.G.; investigation, A.D.; resources, A.D., data curation, T.G., A.D.; writing—original draft preparation, A.D., T.G.; writing—review and editing, all; visualization, T.G.; medical supervision, C.P.; project administration, T.W., Z.S.; funding acquisition, A.D. All authors have read and agreed to the final version of the manuscript.

INFORMED CONSENT STATEMENT

Informed consent was obtained from all subjects involved in this study.

ETHICS COMMITTEE APPROVAL

This study was conducted according to the guidelines of the Declaration of Helsinki and approved by the Ethics Committee of the Canton of Bern (no. 2022-00767).

REFERENCES

1. Huijzer R, de Jonge P, Blaauw FJ, Baatenburg de Jong M, de Wit A, Den Hartigh RJR. Predicting special forces dropout via explainable machine learning. *Eur J Sport Sci*. 2024;24:1564–72. https://doi.org/10.1002/ejsc.12162
2. Pattyn N, Van Cutsem J, Lacroix E, et al. Lessons from special forces operators for elite team sports training: how to make the whole greater than the sum of the parts. *Front Sports Act Living*. 2022;4:780767. https://doi.org/10.3389/fspor.2022.780767
3. McDonald DG, Norton JP, Hodgdon JA. Training success in U.S. Navy special forces. *Aviat Space Environ Med*. 1990;61:548–54.
4. Gayton SD, Kehoe EJ. A prospective study of character strengths as predictors of selection into the Australian army special force. *Mil Med*. 2015;180:151–7. https://doi.org/10.7205/milmed-d-14-00181
5. Picano JJ, Roland RR, Williams TJ, Bartone PT. Assessment of elite operational personnel In: Bowles SV, Bartone PT, eds. *Handbook of Military Psychology: Clinical and Organizational Practice*. Springer International Publishing; 2017:277–289.
6. Schweizer Armee KA. *Rekrutierungs- Und Drop-out-Rate Der Gren RS/ Spez Kräfte RS Und Der Fsch Aufkl RS Vom 2019 Bis 2023*. In: Veil FLM, editor. 2023.
7. Benedict TM, Grier TL, Roy TC, Toussaint MN, Jones BH. Demographic, psychosocial, and physical fitness predictors of successful graduation from U.S. Army ranger school. *Military Psychology: The Official Journal of the Division of Military Psychology, American Psychological Association*. 2023;35:180–91. https://doi.org/10.1080/08995605.2022.2094174
8. Army National Guard Blog Admin. *SFAS Selection Rate Statistics*. Army SF National Guard Blog; 2021.
9. Kimmons S. OPAT Reducing trainee attrition, avoiding millions in wasted training dollars, officials say. *Army News Service*. 2018. Accessed June 18, 2025. https://www.army.mil/article/207956/opat_reducing_trainee_attrition_avoiding_millions_in_wasted_training_dollars_officials_say
10. Sammito S, Hadzic V, Karakolis T, et al. Risk factors for musculoskeletal injuries in the military: a qualitative systematic review of the literature from the past two decades and a new prioritizing injury model. *Mil Med Res*. 2021;8:66. https://doi.org/10.1186/s40779-021-00357-w
11. Knapik JJ, Farina EK, Ramirez CB, Pasiakos SM, McClung JP, Lieberman HR. Medical encounters during the United States army special forces assessment and selection course. *Mil Med*. 2019;184:e337–43. https://doi.org/10.1093/milmed/usz056
12. Stannard J, Fortington L. Musculoskeletal injury in military special operations forces: a systematic review. *BMJ Mil Health*. 2021;167:255–65. https://doi.org/10.1136/bmjmilitary-2020-001692
13. Linderman JK, O'Hara R, Ordway J. Effect of special operations training on testosterone, lean body mass, and strength and the potential for therapeutic testosterone replacement: a review of the literature. *Journal of Special Operations Medicine: A Peer Reviewed Journal for SOF Medical Professionals. Spring*. 2020;20:94–100. https://doi.org/10.55460/fpeq-kdm2
14. Wenger JW, Schaefer AG, Strong B, Buryk P, Waugaman A. *Assessment and selection for U.S. Air Force special warfare: volume 1, defining attributes and designing rater training* 2021. https://www.rand.org/pubs/research_reports/RRA420-1.html
15. Haydu LE, Hogan KK, Merseal C, et al. Impact of an integrated human performance support group: evaluation of air force special warfare candidate training and musculoskeletal injury outcomes over eight fiscal years. *Mil Med*. 2025;190:e251–8. https://doi.org/10.1093/milmed/usae354
16. Krueger TC, Robson S, Walsh M, et al. *Assessment and Selection for U.S. Air Force Special Warfare: Vol. 2, Evaluating Objectives and considering Improvements*. 2024. https://www.rand.org/pubs/research_reports/RRA1199-1.html
17. Fisher R, Esparza S, Nye NS, et al. Outcomes of embedded athletic training services within United States air force basic military training. *J Athl Train*. 2021;56:134–40. https://doi.org/10.4085/1062-6050-0498.19
18. Weatherwax RM, Harris NK, Kilding AE, Dalleck LC. Incidence of V˙O-2max responders to personalized versus standardized exercise prescription. *Med Sci Sports Exerc*. 2019;51:681–91. https://doi.org/10.1249/mss.0000000000001842
19. Bouchard C. Genomic predictors of trainability. *Exp Physiol*. 2012;97:347–52. https://doi.org/10.1113/expphysiol.2011.058735
20. Flück M, Hoppeler H. Molecular basis of skeletal muscle plasticity—from gene to form and function. *Rev Physiol Biochem Pharmacol*. 2003;146:159–216. https://doi.org/10.1007/s10254-002-0004-7
21. Gasser B, Fitze D, Franchi M, et al. The cardiovascular response to interval exercise is modified by the contraction type and training in proportion to metabolic stress of recruited muscle groups. *Sensors*. 2020;21:173. https://doi.org/10.3390/s21010173
22. Ross R, Goodpaster BH, Koch LG, et al. Precision exercise medicine: understanding exercise response variability. *Br J Sports Med*. 2019;53:1141–53. https://doi.org/10.1136/bjsports-2018-100328
23. Bray MS, Hagberg JM, Pérusse L, et al. The human gene map for performance and health-related fitness phenotypes: the 2006-2007 update. *Med Sci Sports Exerc*. 2009;41:35–73. https://doi.org/10.1249/mss.0b013e3181844179
24. Radak Z, Taylor AW. Issues on trainability. *Front Physiol*. 2021;12:790196. https://doi.org/10.3389/fphys.2021.790196
25. Pignanelli C, Christiansen D, Burr JF. Blood flow restriction training and the high-performance athlete: science to application. *J Appl Physiol*. 2021;130:1163–70. https://doi.org/10.1152/japplphysiol.00982.2020
26. Flück M. Diagnostics of endurance performance on the level of gene expression. *Sport-Orthopädie—Sport-Traumatologie—Sports Orthopaedics and Traumatology*. 2013;29:203–13. https://doi.org/10.1016/j.orthtr.2013.07.014
27. Furrer R, Hawley JA, Handschin C. The molecular athlete: exercise physiology from mechanisms to medals. *Physiol Rev*. 2023;103:1693–787. https://doi.org/10.1152/physrev.00017.2022
28. Pickering C, Kiely J. The development of a personalised training framework: Implementation of emerging technologies for performance. *JFMK*. 2019;4:25. https://doi.org/10.3390/jfmk4020025
29. Cruz A, Gomes D, Verdan C, et al. ACE I/D and AGT Met235Thr polymorphisms distinctly affect biomarker levels and risk of AKI and exertional rhabdomyolysis after intense exercise. *Mil Med*. 2025;190:e1920–8. https://doi.org/10.1093/milmed/usaf041

30. Jones N, Kiely J, Suraci B, et al. A genetic-based algorithm for personalized resistance training. *Biol Sport*. 2016;33:117–26. https://doi.org/10.5604/20831862.1198210
31. Scanlan TK, Carpenter PJ, Simons JP, Schmidt GW, Keeler B. An introduction to the sport commitment model. *Journal of Sport and Exercise Psychology*. 1993;15:1–15. https://doi.org/10.1123/jsep.15.1.1
32. Marsh HW, Craven RG. Reciprocal effects of self-concept and performance from a multidimensional perspective: beyond seductive pleasure and unidimensional perspectives. *Perspectives on Psychological Science*. 2006;1:133–63. https://doi.org/10.1111/j.1745-6916.2006.00010.x
33. Dössegger A, Gsponer T, Flück M, et al. Fitness profile and training of special operation forces: a comparison with sports athletes. *Front Sports Act Living*. 2025;7:1594714. https://doi.org/10.3389/fspor.2025.1594714
34. Neuenschwander B, Capkun-Niggli G, Branson M, Spiegelhalter DJ. Summarizing historical information on controls in clinical trials. *Clin Trials*. 2010;7:5–18. https://doi.org/10.1177/1740774509356002
35. Neuenschwander B, Weber S, Schmidli H, O'Hagan A. Predictively consistent prior effective sample sizes. *Biometrics*. 2020;76:578–87. https://doi.org/10.1111/biom.13252
36. Harris PA, Taylor R, Minor BL, et al.; REDCap Consortium. The REDCap consortium: building an international community of software platform partners. *J Biomed Informat. 2019/07/01*. 2019;95:103208. https://doi.org/10.1016/j.jbi.2019.103208
37. Bandura A. Self-efficacy: toward a unifying theory of behavioral change. *Psychol Rev*. 1977;84:191–215. https://doi.org/10.1037//0033-295x.84.2.191
38. den Hartigh RJR, Huijzer R, Blaauw FJ, de Wit A, de Jonge P. Early identification of dropouts during the special forces selection program. *Sci Rep*. 2025;15:3242. https://doi.org/10.1038/s41598-025-87604-5
39. Stallings J, Fisher A. Modeling attrition to inform training pipeline decisions for U.S. Army Special Forces. 2019. CAA-2019060100. https://apps.dtic.mil/sti/pdfs/AD1085170.pdf